Denoising Capacity of Implicit Image Representation

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Abstract—Parameterizing an implicit function with neural network is a popoular approach for signal representation. In this paper, we use sinusoidal representation networks (SIREN) to represent an RGB image. We explore the capacity to which fitting a single image with SIREN can be used for denoising, without access to an external dataset that implicitly represent a natural image prior. After exploring several variants and training conditions of SIREN, results show that when fitting SIREN to a single noisy image, SIREN is able to fit the details of the image first before overfitting to fit the noise. This is a similar phenomenon to Deep Image Prior and suggests SIREN's suitability for representing image signals, because to an extent, it has an inductive bias towards noise-free images.

Index Terms—Computational Imaging, Neural Implicit Functions, Signal Representation, Inverse Problems, Denoising

1 Introduction

REAL world signals are conventionally captured and recorded by sensors as discrete-valued functions, because we do not have infinite resolution. For example, a digital image is a discrete-valued function that is piecewise constant at each pixel coordinate, and a piece of recording is a piecewise constant function at each time stamp. However, this is not a faithful representation of the physical world, which is mostly continuous.

Instead of obtaining a discrete-valued function as captured signal, a different view of signal processing attempts to capture the continuous nature of real world signals. In particular, we can view the sensor output as a set of discrete sample points at evenly spaced points in the continuous space. Therefore, we can parameterize a continuous function and make that function fit these discrete sample points. If the parameterization of this continuous function is flexible and make sense at interpolation points to the specific domain of the particular signal, then we have a good continuous approximation of the real world signal at infinite resolution. These continuous approximations are called implicit functions, because they mimic the sensor captured signal where the input is a (continuous) point in the spatial domain, and the output is the n-dimensional value at that point.

Sinusoidal representation networks (SIREN) [1] is one such example of parameterized implicit functions. It uses a neural network with sinusoidal activation functions to parameterize the implicit function. SIREN has been shown to be able to model signals with fine details as well as capture the spatial and temporal derivatives of the implicit function. It has been shown that SIREN is particularly suitable for capturing signals such as natural images or 3D signed distance fields [1].

In this paper, we apply SIREN to fit natural RGB images. We want to examine the extent to which SIREN can be used to denoise an image in a non-data-driven way, by directly fitting to a noisy image. In other words, we hope to extract a clean image during the training of SIREN when its goal is to fit a noisy image. Not only does it lead to a much faster neural denoising method than data-driven approach like Denoising Convolutional Neural Networks (DnCNN) [2], it also examines the existence of inherent inductive bias of the SIREN construction itself. Because image signals from the real world are sampled from a very different distribution to common sensor noise, if SIREN is able to decouple meaningful lower-frequency signals from high-frequency noise, it suggests that SIREN may be particularly suitable for certain domains of signals.

From training a few variants of SIREN on different objectives, we have found that much like convolutional neural networks (CNN) that have an inductive bias for smooth, natural-looking images [3], SIREN also favors natural-looking images. However, SIREN is much more sensitive to high-frequency details than CNNs, which can still be differentiated from extremely high-frequency noise with a simple filtering method. In general, even without explicitly specifying a denoising objective, we have found SIREN to exhibit comparative denoising performance to a similar setup using CNNs [3], as well as to the dedicated data-driven denoiser of DnCNN [2] and non-data-driven denoiser of BM3D [4]. This shows a promising application of SIREN as general purpose parameterized implicit function for 2D image signals that both preserves high-fidelity details and is robust to random noise.

The code for this project is available at: https://github.com/LeoZDong/siren_denoise.

2 Related Work

There are many works in designing implicit functions for representing 3D signals. Occupancy Networks [5] uses neural networks to parameterize a continous occupancy grid, where for each 3D coordinate, the network outputs the probability of occupancy. When trained on the occupancy of a 3D shape, the p-crossing of an occupancy network extracts the level set of the shape with confidence p, where $p \in [0, 1]$. Similarly, DeepSDF [6] uses neural networks to

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parameterize a signed distance function, where each for each 3D coordinate, it outputs the signed distance to the surface of the shape of interest. These methods are similar to SIREN except they typically use ReLU activation. Without periodic activation functions, they are generally not suitable for image signals as shown in [1] because they tend to miss fine details. There are neural implicit functions that use smart activations or parameterizations like SIREN to achieve finer details, such as Fourier Features [7] and Spline Positional Encoding [8]. In one of the model variants, we add the Spline Positional Encoding to a regular SIREN model, and its details are discussed in the methods section.

They are also a variety of local-based implicit functions. The general idea is that not all parts of the spatial domain contains the same amount of information. So we can learn a higher fidelity model if we learn multiple implicit functions at different scales, dedicating an entire implicit function to focus on a small area with high-frequency details, for example. Local Deep Implicit Functions [9] is a local-based approach for learning signed distance functions, and Local Implicit Image Function [10] is a local-based approach for fitting to natural images. While these methods offer higher fidelity, they often take longer to train and the resulting model is hard to analyze. Because a basic neural implicit function is still the foundational building block for these methods, we decide to make studying the basic neural implicit function itself the focus of this paper.

There are many non-data-driven and data-driven approaches for denoising. Denoising Convolutional Neural Network (DnCNN) [2] is a popular CNN architecture that uses residual networks to learn to denoise. This is a data-driven approach so the network is first trained on a large dataset of clean-noisy image pairs, and the network output is optimized to match the clean image while given noisy image as input. After we have a trained network, we can pass any new noisy image to the network as input and the output is the denoised result. Even though DnCNN also uses neural networks, it is a very different approach from SIREN because SIREN is only fit on a single image (i.e. the noisy image we are given) and we do not show it examples of what a "clean" image should look like. BM3D [4] is a state-of-the-art non-learning and non-data-driven approach. It follows the basic idea of the popular nonlocal means [11] by exploring the self-similarity of image patches in an image. For each image patch, BM3D searches for simlar patches in the image, weight them by similarity, and take the weighted average as the combined denoised result of that patch. We use DnCNN to represent a neural data-driven approach and BM3D to represent a classical non-data-driven approach as baselines in the experiments.

Finally, Deep Image Prior [3] explores the same idea in Convolutional Neural Networks (CNN). Deep Image Prior tries to fit a bottlenecked CNN model (intermediate dimension is lower than input and output dimensions) to a noisy or corrupted image, and observes that CNNs are naturally biased towards fitting the geometric information of the image but not the noise information of the image, so we can get a clean image if we just early-stop the training that fits a noisy image. While the observation is similar for SIREN, we see that SIREN is more comfortable at fitting high-frequency information so the early-stopped result has more visible noise than CNN. But because the continuous function that SIREN learns still decouples noise information from geometry information to an extent, we can significantly remove the noise if we just average (continuous) neighboring points because the noise information is not baked into the interpolated grid points. We will discuss the comparison with Deep Image Prior more closely in the results section.

3 Proposed Method

We first describe the basic formulation of SIREN, and then the Spline Positional Encoding (SPE) that we add to SIREN as a variant. Then we discuss the 3 training conditions that we test SIREN on.

3.1 SIREN Formulation

SIREN parameterizes a neural network Φ that takes an input grid in the spatial domain of the signal and outputs the signal intensities at each grid point. Specifically, Φ maps

$$\Phi(\mathbf{x}) = \mathbf{W}_n(\phi_{n-1} \circ \phi_{n-2} \circ \cdots \circ \phi_0)(\mathbf{x}) + \mathbf{b_n}, \tag{1}$$

where $\mathbf{x} = \{\mathbf{x}_1, ..., \mathbf{x}_N\}$ is a set of grid coordinates. Because we are fitting to 2D images, $\mathbf{x}$ is a set of 2D coordinates. Each ϕ_i is a layer of the network that maps a single grid coordinate $\mathbf{x_j}$ to

$$\phi_i(\mathbf{x}_j) = \sin(\mathbf{W}_i\mathbf{x}_i + \mathbf{b_i}). \tag{2}$$

The trainable parameters of SIREN are $\mathbf{W}_i$ and $\mathbf{b_i}$ for $i = 0, 1, ..., n$.

3.2 SPE Formulation

We can also apply spline positional encoding (SPE) [8] to the each input grid coordinate $\mathbf{x_i}$ as follows: we first select m random unit directions $D_1, ..., D_m$, and project the 2D coordiante $\mathbf{x_i}$ to a 1D coorindate $x_i := \langle \mathbf{x_i}, D_i \rangle$. Then the encoding of $\mathbf{x_i}$ is

$$S(\mathbf{x_i}) = [\psi_1(x_1), ..., \psi_m(x_m)]. \tag{3}$$

Each ψ_i is a parametric spline function defined by

$$\psi(x) = \sum_{i=0}^{K} \tilde{W}_i B_{\delta, c_i}(x) \tag{4}$$

$$B_{\delta, c_i}(x) := B(\frac{x - c_i}{\delta}), \tag{5}$$

and B is a standard n-th degree spline basis function. The degree of B is fixed. The trainable parameters of SPE are thus $\tilde{W}_i, \delta, c_i$ for $i = 0, 1, ..., K$. Therefore, if we apply SPE to SIREN, the final model becomes

$$\Phi(\mathbf{x}) = \mathbf{W}_n(\phi_{n-1} \circ \phi_{n-2} \circ \cdots \circ \phi_0 \circ S)(\mathbf{x}) + \mathbf{b_n}. \tag{6}$$

3.3 Training Conditions

We are given a corrupted image $\hat{I}$ that is obtained from $\hat{I} = I + \epsilon$, where $\epsilon \sim \mathcal{N}(0, \sigma)$ and I is unknown. We model $\hat{I}$ as a continuous-valued function, $\hat{I} : [-1, 1]^2 \to [0, 1]^3$, that maps each continuous coordinate to a 3-channel RGB value. We set $\Omega := [0, 1]^2$ to be the domain of $\hat{I}$. But the captured image only obtains sample points of $\hat{I}$ at a discrete set of grid points, G, where the points in G is evenly spaced on the grid $[-1,1]^2$. Because we are using 256×256 resolution images, $|G| = 256^2$.

We test SIREN under three conditions. First is fitting a basic SIREN model directly to a noisy image. We call it the fidelity model because it only minimizes the data fidelity loss

$$\mathcal{L}_{\text{fidelity}} = \int_{\Omega} \|\Phi(\mathbf{x}) - \hat{I}(\mathbf{x})\|_2 d\mathbf{x}.$$

Next, we add total variation (TV) regularization to SIREN that enforces SIREN to have sparse gradients. Because the sinusoidal activation of SIREN is continuously differentiable at every point in the domain, we can define

$$\mathcal{L}_{\text{tv}} = \int_{\Omega} \|\Phi(\mathbf{x}) - \hat{I}(\mathbf{x})\|_2 + \kappa \|\nabla_{\mathbf{x}} \Phi(\mathbf{x})\|_1 d\mathbf{x} \tag{7}$$

as the objective for the TV model. κ is the relative weight of TV regularization and is a hyperparameter.

Finally, we apply SPE to SIREN's input coordinate and train with the objective

$$\mathcal{L}_{\text{spe}} = \int_{\Omega} \|\Phi(S(\mathbf{x})) - \hat{I}(\mathbf{x})\|_2 d\mathbf{x}, \tag{8}$$

which we call the SPE model. In all three conditions, the integral over the domain Ω is approximated by the sum over grid points in G that we have observations for $\{\hat{I}(g) \mid g \in G\}$.

3.4 Continuous Bilateral Filtering

Because we have learned a continuous functional representation of Φ for an image, we can be a lot more flexible at applying classical denoising techniques. Here we apply a continuous type of bilateral filtering. Specifically, we randomly perturb the input grid G for f times, and for each perturbed grid $\hat{G}^{(j)}$, we define the position weight as

$$W_{\text{pos}}(G, \hat{G}^{(j)})_k = \exp\left(-\frac{\|G_k - \hat{G}_k^{(j)}\|}{2\sigma_p^2}\right) \tag{9}$$

and the intensity weight as

$$W_{\text{int}}(\Phi(G), \Phi(\hat{G}^{(j)}))_k = \exp\left(-\frac{\|\Phi(G)_k - \Phi(\hat{G}^{(j)})_k\|}{2\sigma_i^2}\right). \tag{10}$$

Then the filterd image is

$$\frac{1}{Z}\sum_{j=1}^{m} W_{\text{pos}}(G, \hat{G}^{(j)}) W_{\text{int}}(\Phi(G), \Phi(\hat{G}^{(j)})) \hat{G}^{(j)}, \tag{11}$$

where Z is the normalization factor. This is a much more flexible filtering method because we essentially have infinite resolution and can query points just a little bit next to the input grid. The rationale behind this continuous bilateral filtering is that while SIREN may still fit some noise in the learned Φ, because noise is extremely high frequency and has no meaningful neighborhood structure, the interpolated neighboring points $\hat{G}_k^{(j)}$ next to a point corrupted by noise at G_k most likely do not contain the same noise, so averaging can filter the noise information out.

3.5 Evaluation Metric

We evaluate the denoised image using Peak Signal-to-Noise Ratio (PSNR) and Learned Perceptual Image Patch Similarity (LPIPS). PSNR directly compares the input image with the target image in terms of normalized squared difference. On the other hand, LPIPS compares the perceptual simiarity to the target image that is more similar to how humans perceive natural images to be similar. For PSNR, higher is better. For LPIPS, lower is better.

4 Experimental Results

We discuss some interesting experimental results when training variants of SIREN to fit the noisy image directly.

4.1 SIREN Training has Two Phases

When fitting SIREN to the noisy image, we calculate two metrics of LPIPS, `lpips_clean` and `lpips_noisy`. `lpips_clean` measures the LPIPS between the current model output and the clean (unknown) image, and `lpips_noisy` measures the LPIPS between the current model output and the noisy (known) image that is the training target. We do the same for PSNR.

We show the training plots of the fidelity model, TV model, and SPE model in figures 1 and 2, for LPIPS and PSNR respectively. We see that while the metric of the model output compared to the target (noisy) image keeps improving, the model changes from the first phase of getting increasingly similar to the clean image to a second phase of getting less similar to the clean image. For the fidelity model, this happens at around iteration 600. Therefore, we can visualize the fidelity model output on different points of the training curve. Figure 3 shows the model output at step 60, 600, and 1180 respectively. We see that from step 60 to 600, the biggest improvement is that the model learns a much more detailed geometry of the image. From step 600 to 1180, the details of the geometry is almost the same, but step 1180 just has a lot more noise. This means that during phase 1 of SIREN model training, the model is fitting both the geometry and some high-frequency noise, while during phase 2 of training, the model is only fitting the rest of the noise.

4.2 SPE and TV as Regularization

Next, we examine result of the TV and SPE model. Figure 4 and 5 show the LPIPS and PSNR training curves. As a popular natural image regularization method, TV is able to push `LPIPS_clean` always low, although there is still a two-phase phenomenon for the PSNR metric. When we visualize the gradient of the fidelity and TV models at step 1180 in figure 6, we see that this is because TV successfully constrains the gradient to be sparse. However, choosing the right κ is a bit tricky: if we make κ too large, the reconstructed geometry can be too soft compared to the clean image, missing a lot of high-frequency details that are meaningful.

On the other hand, SPE shows an even stronger regularization power than TV because it gets rid of the two-phase phenomenon in both LPIPS and PSNR. SPE's regularization is in the implicit model definition that the input coordinates

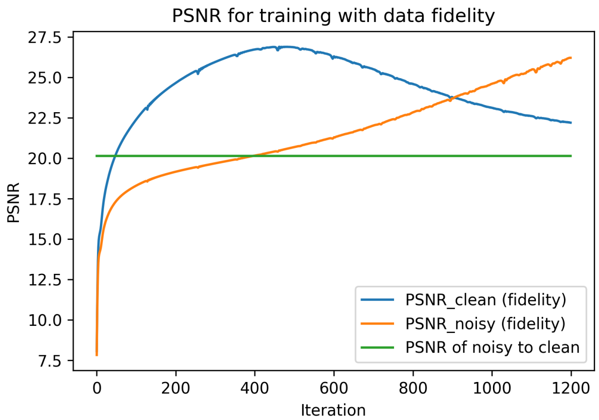

Fig. 1. PSNR training curves for fidelity model. Higher is better. Green curve plots the baseline PSNR of the input noisy image with the clean image.

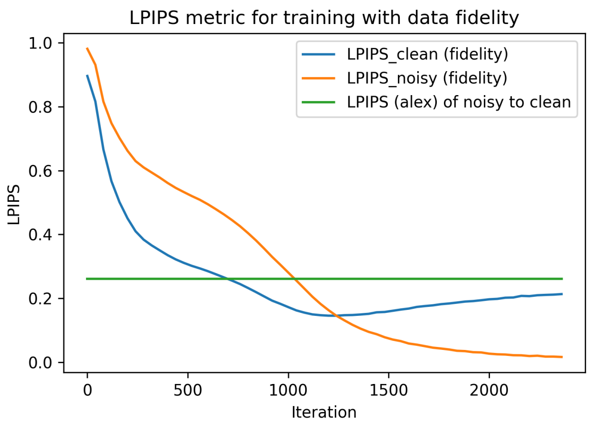

Fig. 2. LPIPS training curve for fidelity model. Lower is better. Green curve plots the baseline LPIPS of the input noisy image with the clean image. Correction: x-axis should e from 0 to 1200, not 0 to 2400.

are represented as projected encodings on spline functions. SPE ends up being an "easier" regularizer than TV because there is no hyperparameter and we can get a slightly better metric. We can also see that SPE regularization is stronger because it takes longer for TV to converge, which could be a downside to using SPE.

4.3 Continuous Bilateral Filtering Removes Noise

We apply the continuous bilateral filtering to the fidelity model. Figure 7 shows comparison between the originaly early-stopped model output and the continous bilateral filtered model output. We see that both qualitatively and quantitatively, with PSNR increasing from 21.28 to 27.01 and LPIPS decreasing from 0.15 to 0.102, the continuous bilateral filtering is very effective at removing noise. While it may require further investigation, one explanation is that even though SIREN is good at fitting high-frequency signal and thus includes visible noise in the output, the extremely high-frequency noise does not have good neighborhood structure, so the noise signal cannot bleed into interpolated neighors. In other words, say a particular point in the input grid G_k is corrupted by some noise ϵ_k. Because the neighbors of G_k in G have very different noise, the interpolated continuous neighborhood of G_k does not show consistent signals of ϵ_k. Therefore, if we just take a small continuous neighborhood of G_k and return the weighted average, the noise will be filtered out. And because we can apply this filtering at a much smaller scale than traditional bilateral filtering, this does not blur out the geometric details as much.

4.4 Comparison with Baselines

Figures 8 and 9 show the quantitaive comparisons of LPIPS and PSNR between SIREN variants and the baseline methods of Deep Image Prior, BM3D, and DnCNN. Note that because DnCNN is a data-driven approach, we expect it to have better performance than all the other non-data-driven approaches, so it is presented as an expected upper bound performance. Figures 10 shows the qualitative results.

We see that among the SIREN variants, TV has the worst performance. This might be because the TV regularization term weight κ was not extensively tuned, so the results are overly smoothing the image and leave out a lot of geometric details. Indeed, we see that the TV result is qualitatively oversmoothed. This shows one downside of using TV because the output is very sensitive to TV weight. Fidelity model shows better metric than the SPE model, although visually, the outputs of the two models are very similar. Even though the fidelity model trains faster than SPE and its early-stopped result is good, one advantage of SPE is that it does not require manual inspection to decide which step to early stop at, because SPE regularizes the training where we do not overfit the noise.

The continuous bilateral filtered SIREN fidelity model is a significant improvement, which makes SIREN very comparable with all the three baseline methods, including BM3D. However, we do see that compared to DnCNN and Deep Image Prior, the filtered SIREN is lacking some details, which may be due to the specific choice of the filtering parameters.

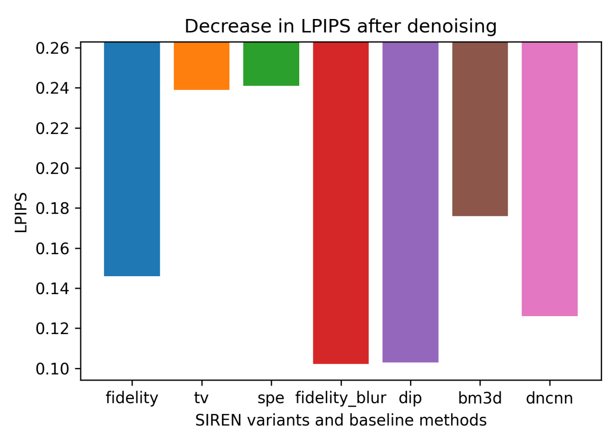

Fig. 8. Comparison of LPIPS between SIREN variants and baseline methods.

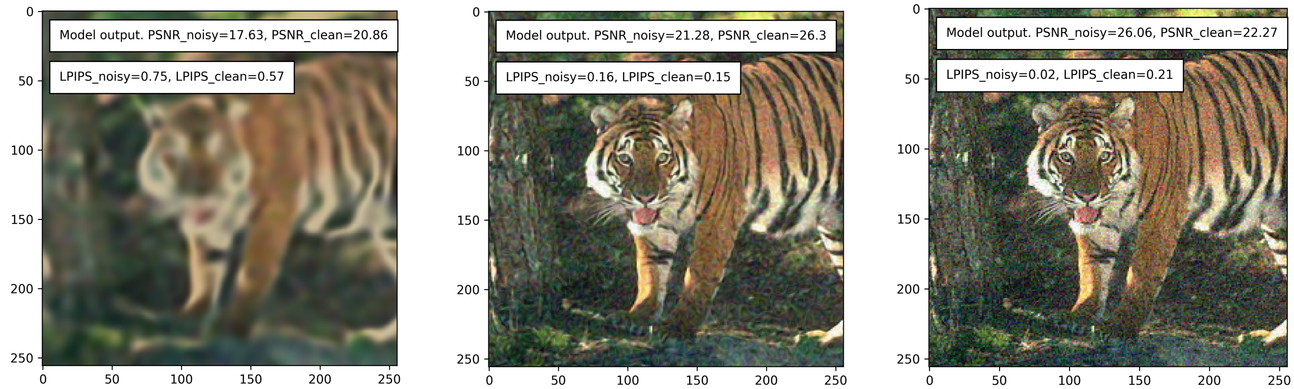

Fig. 3. Fidelity model output at steps 60, 600, and 1180 respectively.

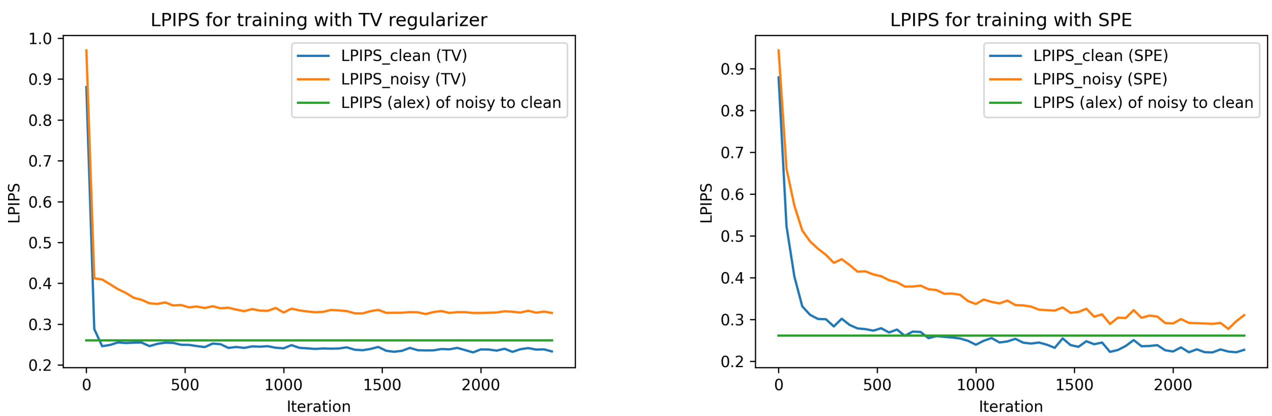

Fig. 4. LPIPS training curves for TV and SPE model. Lower is better.

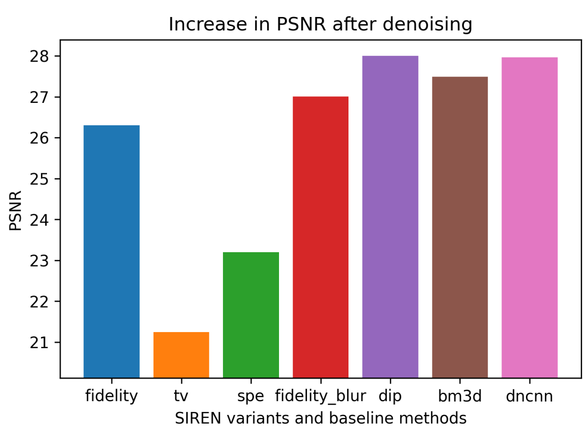

Fig. 9. Comparison of PSNR between SIREN variants and baseline methods.

5 Conclusion

In this paper, we explore the denoising capacity of SIREN and its variants. We see that SIREN exhibits a similar behavior as CNN that biases towards fitting high-frequency noise last, making it able to denoise without access to a large dataset and by simply fitting the model to a noisy image. Both TV and SPE are sensible regularization methods, although TV is sensitive to tuning the regularization weight. We also see that SIREN is flexible in deploying continuous analogs of classical pixel-based denoising methods, which can significantly improve the result. The resulting denoised output of SIREN is comparable to non-data-driven methods of BM3D and Deep Image Prior, as well as data-driven method of DnCNN.

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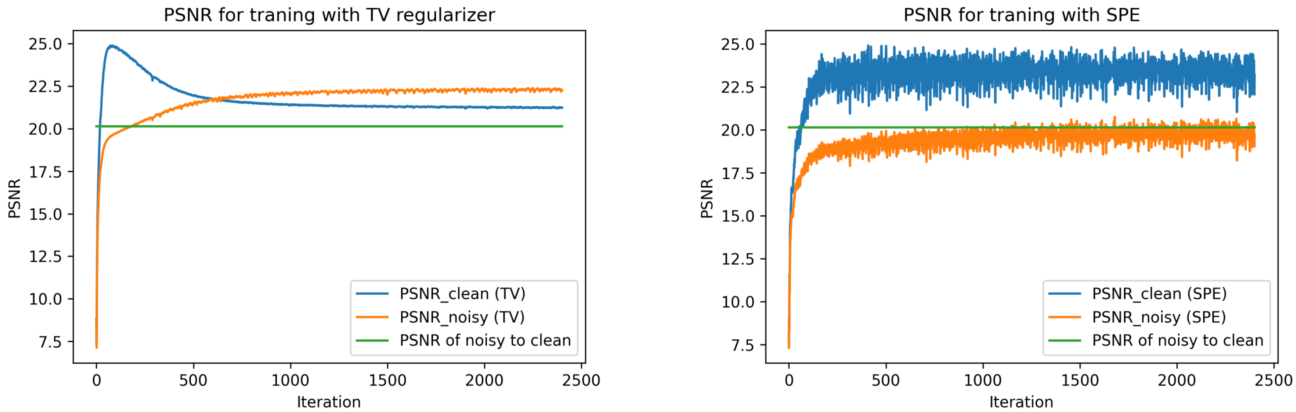

Fig. 5. PSNR training curves for TV and SPE model. Higher is better.

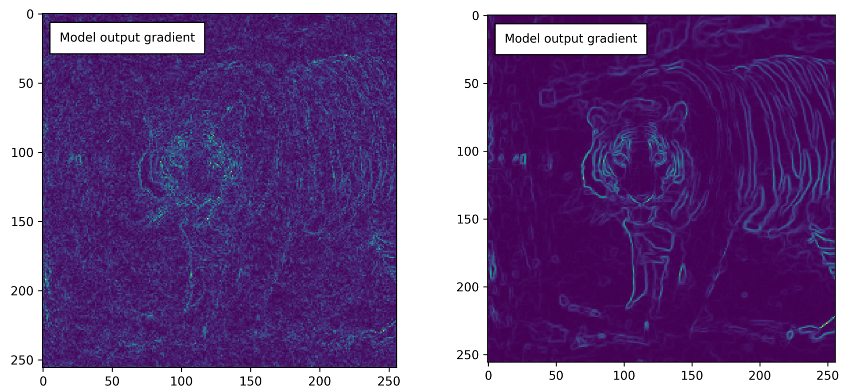

Fig. 6. Fidelity model (left) and TV model (right) gradient at step 1180.

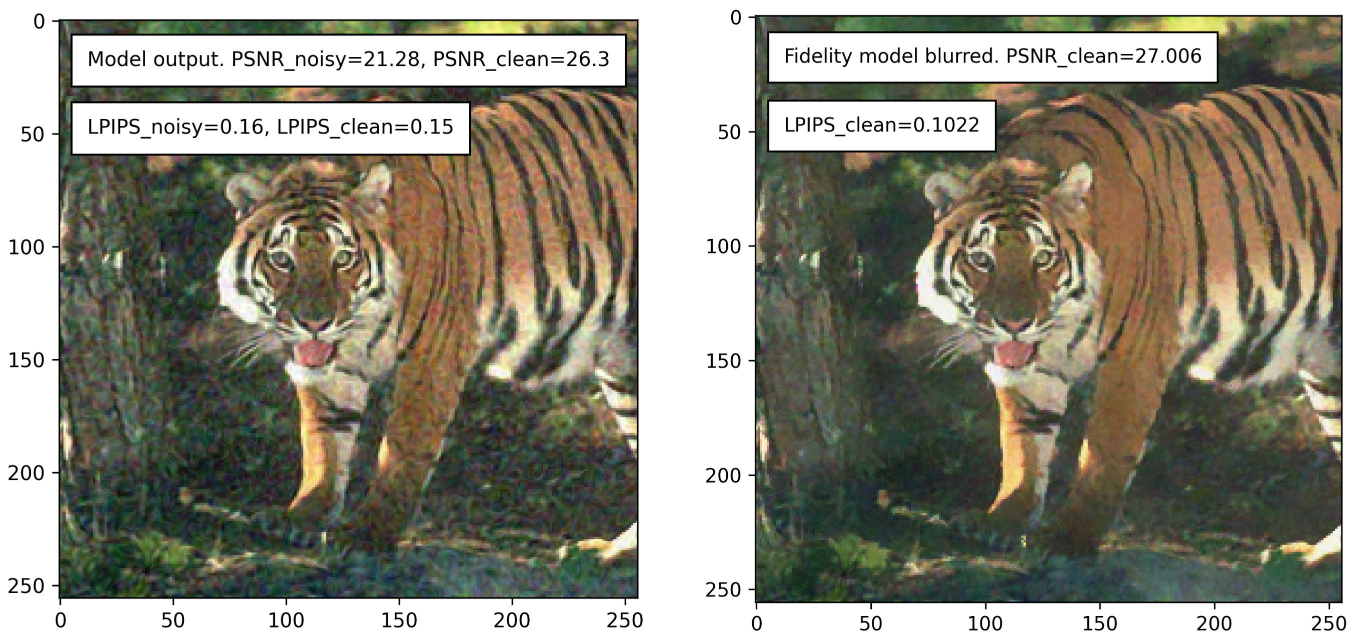

Fig. 7. Early-stopped fidelity model output (left) and continous bilateral filtered model output (right).

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Fig. 10. Qualitative comparison of SIREN variants and baselines.

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